

RAPPORT DE STAGE DE RECHERCHE

Deeper or wider ? A first answer under the NTK perspective

An optimization perspective on Neural Network design

RAPPORT NON CONFIDENTIEL

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ABSTRACT

In the past few years, research has focused on understanding the fundamental mechanisms governing learning dynamics in deep neural networks. While significant progress has been made through the **Neural Tangent Kernel (NTK)** formalism, it remains unclear how to choose the right architecture for a given problem. In the tryptical approximation/generalization/optimization trilemma, we focus on the optimization aspect, the less understood one. This report presents a comprehensive answer to this question through the lens of **NTK theory** in the context of **Sobolev training**.

We establish a novel theoretical framework and our key finding demonstrates that the effectiveness of learning depends critically on **sampling conditions**, specifically when using **Chebyshev points**. For general sampling schemes, optimal learning requires **explicit derivative information** for effective training.

We derive **explicit theoretical and numerical bounds** for key neural network quantities establishing **precise eigenvalue bounds** for **NTK** matrices and ongoing **quantitative convergence rates**.

Our **spectral analysis** covers **fully connected networks (FCNN)**, **Multi-Layer Multi-component networks (MMNN)**, and **transformers**, revealing architecture-specific scaling with depth, width, and input dimension. Through **finite-width corrections**, we provide **explicit error bounds** and foundations for ongoing **convergence guarantees** for gradient descent.

Experimental validation confirms our theoretical predictions across all architectures. Fully reproducible code is available at github.com/janisaiad/DeNN-NTK and github.com/janisaiad/MMN.

RÉSUMÉ

Au cours des dernières années, la recherche s'est concentrée sur la compréhension des mécanismes fondamentaux régissant la dynamique d'apprentissage dans les réseaux de neurones profonds. Bien que des progrès significatifs aient été réalisés grâce au formalisme du **Neural Tangent Kernel (NTK)**, il reste peu clair comment choisir la bonne architecture pour un problème donné. Dans le trilemme classique approximation/généralisation/optimisation, nous nous concentrons sur l'aspect optimisation, le moins bien compris. Ce rapport présente une réponse complète à cette question à travers le prisme de la **théorie du NTK** dans le contexte de **l'entraînement de Sobolev**.

Nous établissons un nouveau cadre théorique et notre découverte clé démontre que l'efficacité de l'apprentissage dépend de manière critique des **conditions d'échantillonnage**, spécifiquement lors de l'utilisation de **points de Chebyshev**. Pour les schémas d'échantillonnage généraux, l'apprentissage optimal nécessite une **information explicite sur les dérivées** pour un entraînement efficace.

Nous dérivons des **bornes théoriques et numériques explicites** pour les quantités clés des réseaux de neurones, établissant des **bornes précises sur les valeurs propres** pour les matrices **NTK** et des **taux de convergence quantitatifs** en cours.

Notre **analyse spectrale** couvre les **réseaux entièrement connectés (FCNN)**, les **réseaux multi-couches multi-composants (MMNN)**, et les **transformers**, révélant une mise à l'échelle spécifique à l'architecture avec la profondeur, la largeur et la dimension d'entrée. À travers les **corrections de largeur finie**, nous fournissons des **bornes d'erreur explicites** et des fondations pour les **garanties de convergence** en cours pour la descente de gradient.

La validation expérimentale confirme nos prédictions théoriques à travers toutes les architectures. Un code entièrement reproductible est disponible sur github.com/janisaiad/DeNN-NTK et github.com/janisaiad/MMN.

INTRODUCTION

The study of learning dynamics in deep neural networks constitutes a fundamental research area. Our approach focuses on the desingularization of Sobolev training by decomposing the problem into two distinct contributions: an independent neural tangent kernel (NTK) and a Sobolev contribution. This decomposition relies on the commutation between integral operators and the fractional Laplace operator.

Key Discovery: We establish that this commutation property holds *exactly* only when using **Chebyshev points** (or equivalent optimal interpolation nodes) for sampling. For general sampling points, commutation breaks, necessitating explicit derivative information for effective learning.

Our analysis continues with the study of NTK preconditioning through a freeze weights method, illustrated by neural networks factorizing low-rank weight matrices. This approach is supported by a rigorous approximation theorem. For classical feed-forward networks (FCNN), we develop a finite-width correction that proves particularly fruitful. Finally, we explore Transformers as an alternative solution, offering a new learning paradigm.

Across these different architectures, we study feature learning regimes and their scaling with network width. Our results rely on statistical physics tools to characterize model convergence and asymptotic behavior. This analysis enables a better understanding of finite-width corrections [[seleznova2022ntk_beyond](#)] and their implications for the practical optimization of deep neural networks.

The end of this research establishes fundamental connections between neural network dynamics and mathematical physics, advancing theoretical understanding of eigenvalue bounds for neural tangent kernels across multiple architectures. We bridge classical optimization theory with advanced mathematical physics frameworks, leveraging tools from quantum field theory and random matrix theory to provide new insights into neural network behavior.

Our contribution to physics-informed deep learning theory applies techniques from statistical mechanics and random matrix ensembles, offering a mathematical physics perspective that complements existing approaches. This interdisciplinary methodology demonstrates how theoretical physics concepts illuminate fundamental machine learning questions, establishing new paradigms for understanding complex neural architectures.

Extensive experimental validation across feed-forward networks, Multi-Layer Multi-component networks, and transformers confirms our theoretical predictions while revealing phenomena that merit further investigation at the intersection of deep learning and mathematical physics.

ACKNOWLEDGEMENTS AND SCOPE

I would like to thank all those who have helped and guided me during this research work.

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The working together and interesting discussions with other researchers have been important for developing the ideas shown here. This work shows the power of working together in research and the generous sharing of knowledge in the scientific community.

SCOPE AND INFORMAL WORDS

We have spent significant time to make sure this report can be understood by many readers while staying clear and complete about our results. We avoid unnecessary details or extra explanations that do not help understand our methods and findings. Every statement is supported and our assumptions are clearly stated. The references are complete and honest about current research, noting that this work is very recent with the latest advances from **August 2025**. To enhance readability and ensure readers have the most enjoyable experience possible, we have strategically incorporated **colors**, I hope the reader will enjoy the reading experience.

This work is still ongoing, supervised by the aforementioned professors. I have never been so happy to work on a subject, particularly because there exists a huge theoretical gap to fill in this emerging field that bridges physics and deep learning, especially **random matrix theory**, **statistical physics** and **quantum field theory**, this will be tackled in the oral presentation.

This report required **1000 pdflatex compilations** - totally worth it!

NOTATIONS

Mathematical Notation Reference

BASIC MATHEMATICAL SYMBOLS

Symbol	Definition	Symbol	Definition
$\mathbb{R}$	Set of real numbers	$\mathbb{N}$	Set of natural numbers
$\mathbb{C}$	Set of complex numbers	$\mathbb{S}^{d-1}$	Unit sphere of dimension $d - 1$
$\mathbb{E}[\cdot]$	Mathematical expectation	$\text{Var}(\cdot)$	Variance of a random variable
$(\cdot, \cdot)$	Covariance between two variables	$\mathbb{I}\{\cdot\}$	Indicator function
dx	Differential element	$\ \cdot\ $	Euclidean norm
$\langle \cdot, \cdot \rangle$	Inner product	$\mathcal{O}(\cdot)$ and $\tilde{\Omega}(\cdot)$	Landau notation

NEURAL NETWORK AND ACTIVATION FUNCTIONS

Symbol	Definition	Symbol	Definition
σ	Activation function	$f(\mathbf{x}; \boldsymbol{\theta})$	Neural network function
σ	General variance denomination	L	Number of layers
$decomposing_k$ (MMNN)	Width of layer k	n_k (FCNN)	Width of layer k

NEURAL TANGENT KERNEL (NTK)

Symbol	Definition	Symbol	Definition
$\mathbf{K}$	Discrete NTK matrix	$\mathbf{K}^\infty$	Infinite-width NTK operator
$K^L(\mathbf{x}_i, \mathbf{x}_j)$	NTK for L -layer network	$\mu_\ell^{(K)}$	NTK eigenvalues at degree ℓ
$\varrho(\rho)$	Correlation propagation function	$\rho_1(\mathbf{x}, \mathbf{x}')$	Initial correlation

Mathematical Notation Reference

SOBOLEV SPACES AND TRAINING

Symbol	Definition	Symbol	Definition
$H^s(\Omega)$	Sobolev space of order s	$\mathbf{P}_s$	Discrete Sobolev operator matrix
P_s	Continuous Sobolev operator	$\mathcal{L}_{\text{Sobolev}}(f)$	Sobolev training loss function
D^k	k -th derivative operator	$\ f\ _{H^s}$	Sobolev norm of order s
$(I + (-\Delta)^{1/2})^s$	Fractional Laplacian operator	$\Delta_{\mathbb{S}^{d-1}}$	Laplace-Beltrami operator

SPHERICAL HARMONICS AND FOURIER ANALYSIS

Symbol	Definition	Symbol	Definition
$Y_{\ell,p}$	Spherical harmonics	$N(d, \ell)$	Multiplicity of harmonics
$\hat{f}(\xi)$	Fourier transform of f	ξ	Frequency variable
$\sigma(x)$	Surface measure on sphere	ℓ	Degree index
p	Order index	∇	Gradient operator

EIGENVALUES AND SPECTRAL ANALYSIS

Symbol	Definition	Symbol	Definition
λ_ℓ	Eigenvalues at degree ℓ	$\lambda_{\min}(A)$	Minimum eigenvalue
$\lambda_{\max}(A)$	Maximum eigenvalue	$\mu_\ell^{(P_s)}$	Sobolev eigenvalues
$C(d, L)$	Constants depending on d, L	$\Gamma(\cdot)$	Gamma function
$P_\ell^{(\alpha)}(t)$	Gegenbauer polynomials	$\tilde{K}_L^\infty(t)$	Normalized NTK function

MATRIX OPERATIONS AND TRAINING

Symbol	Definition	Symbol	Definition
A^T	Matrix transpose	$\ A\ _{op}$	Operator norm
$\ A\ _F$	Frobenius norm	$A \circ B$	Hadamard product
$\mathcal{T}_s$	Composite learning operator	$a_{\ell,p}$	Sampling vectors
η	Learning rate	$f_t(x)$	Network at time t
y_i	Target output	n	Number of samples
d	Input dimension	$\frac{s}{8}$	Sobolev order

1

STATEMENT OF THE PROBLEM

1.0.1 • THE SOBOLEV TRAINING

Let's suppose that we are given a function f^* and we want to find a function f_θ in a model class $\mathcal{F}$ that approximates f^* in the Sobolev space $H^s(\Omega)$.

Sobolev training [czarnecki2017sobolev, gupte2022sobolev_loss] is an approach that incorporates derivative information into the learning process. For a function f and target function f^* , the Sobolev training loss is defined as:

Sobolev Training Loss

$$\mathcal{L}_{\text{Sobolev}}(f) = \|f - f^*\|_{H^s}^2 = \sum_{p=0}^s \int_{\Omega} |D^p f(x) - D^p f^*(x)|^2 dx \quad (1)$$

where D^p denotes the p -th derivative operator and H^s is the Sobolev space of order s . This formulation leads to several key properties:

The Sobolev training loss has the following properties:

- The loss incorporates both function values and derivatives up to order s
- Training in H^s norm induces smoother function approximation

For neural networks, this translates to augmenting the training data with derivative information:

$$\mathcal{L}_{\text{discrete}}(f) = \sum_{i=1}^n \sum_{p=0}^s |D^p f(x_i) - D^p f^*(x_i)|^2$$

This framework provides a natural connection between neural network training and classical function approximation theory in Sobolev spaces. It's important to note that Sobolev training is mainly used for very smooth functions in general. But the terminology "Sobolev" remains the same and is part of the field slang, even if describing it in a sobolev space is ont fully informative.

1.1 LITERATURE REVIEW: SOBOLEV TRAINING AND NTK EIGENVALUES

1.1.1 • THE INTERSECTION OF SOBOLEV TRAINING AND NTK SPECTRAL ANALYSIS

The convergence of Sobolev training methodologies with Neural Tangent Kernel (NTK) spectral analysis represents one of the most promising theoretical frontiers in modern machine learning. This research line fundamentally addresses how embedding derivative information into learning objectives reshapes the spectral and optimization dynamics of neural networks. The literature spans from foundational works in approximation theory to recent finite-width corrections via diagrammatic methods, forming a coherent but rapidly evolving theoretical framework.

Foundations of Sobolev Training The theoretical origins of Sobolev training lie in classical numerical analysis, where controlling derivatives of approximants was essential for stability and convergence. Czarnecki et al. [czarnecki2017sobolev] formalized this approach for deep learning, demonstrating that penalizing derivative mismatches induces smoother approximations and improved generalization properties. We observe that this foundational work established the principle that incorporating higher-order information fundamentally changes the learning dynamics.

Subsequent theoretical refinements by Gupte et al. [gupte2022sobolev_loss] provided rigorous error bounds, linking Sobolev loss formulations to spectral control of learned functions, particularly in high-frequency regimes. This insight established the principle that the Sobolev order s directly dictates approximation error decay rates. We note that these results created the theoretical foundation for understanding how derivative penalties affect function space properties.

Recent developments extend these ideas in two complementary directions. On the theoretical side, Doumeche [doumeche2023sobolev] demonstrated that Sobolev training fundamentally modifies the NTK operator itself, effectively embedding fractional differential operators into the kernel structure. On the applied side, Townsend et al. [townsend2022sobolev] showed the effectiveness of Sobolev losses in PDE-constrained learning scenarios, where derivative-aware training achieves faster convergence and lower sample complexity compared to standard approaches.

Spectral Properties of Neural Tangent Kernels Parallel to Sobolev training advances, NTK analysis has provided the dominant theoretical framework for studying convergence in wide neural networks. Bach’s foundational works [bach2017breaking, bach2021ntk_eigenvalues] established polynomial eigenvalue decay rates for NTKs, demonstrating that spectral tail behavior governs generalization properties, while the smallest eigenvalue $\lambda_{\min}$ dictates optimization convergence speed.

These fundamental results were refined by Nguyen et al. [nguyen2022ntk_eigenvalues] and Fan & Wang [fan2020spectra], who connected eigenvalue distribution patterns to architectural parameters including depth, width, and random matrix effects. We observe that these works established precise relationships between network structure and spectral properties. More recently, Doumeche [doumeche2023ntk_spectral] introduced concentration inequalities for empirical NTK spectra, ensuring stability of eigenvalue distributions in high-dimensional settings.

Finite-width corrections represent an active research area where Hanin & Nica [hanin_nica_2019], Dyer & Gur-Ari [dyer_gur_ari_2019], and Guillen et al. [guillen_misof_gerken_2025] employed diagrammatic techniques inspired by quantum field theory to describe NTK fluctuations. These approaches bridge the theoretical gap between infinite-width and finite-width regimes, providing practical insights for real neural networks.

Convergence of Sobolev Training and NTK Theory The synthesis of these research directions occurs in works such as Doumeche [doumeche2023sobolev_ntk], which demonstrated that Sobolev training reweights the NTK spectrum through powers of differential operators, accelerating decay of high-frequency eigenmodes. This theoretical result explains empirically observed performance gains in PDE solvers and smooth regression tasks, providing the first rigorous connection between Sobolev norms and kernel spectral properties.

Complementary works extend this analysis to optimization guarantees. Welper [welper2023approximation] studied gradient flow behavior for Sobolev-smooth targets, while recent investigations [ntk2024underparam, ntk2025divergence] examined NTK behavior in under-parameterized and classification settings, where uniform kernel convergence can break down. We note that these studies reveal both the power and limitations of NTK-based theoretical analyses.

Convergence Analysis and Open Challenges The combined theoretical framework highlights a fundamental trade-off in neural network training. We observe that Sobolev training accelerates convergence when

target functions possess sufficient smoothness, but may hinder learning for non-smooth signals. Adaptive strategies that dynamically tune the Sobolev order s based on local function properties represent an active research direction to mitigate this sensitivity.

Moreover, sampling strategies critically shape NTK spectral properties, yet remain poorly understood in practice. The connection between data geometry and kernel eigenstructure, particularly for optimal interpolation nodes such as Chebyshev points [**driscoll2014chebfun**], is emerging as a key theoretical and algorithmic lever. It remains unclear why certain sampling patterns lead to dramatically improved spectral properties in the NTK regime.

Finally, extending Sobolev-aware NTK analysis to modern architectures including convolutional networks and attention mechanisms, as well as to stochastic training dynamics, remains a theoretical frontier. Diagrammatic methods and neural tangent hierarchy approaches [**huang2019dynamicsdeepneuralnetworks**] may provide the appropriate mathematical tools for handling such architectural complexity.

The synthesis of Sobolev training and NTK spectral analysis offers a unified theoretical framework for understanding neural network convergence properties. We observe that derivative-aware training objectives fundamentally reshape kernel spectra, altering both optimization dynamics and generalization behavior. Foundational results from approximation theory, kernel eigenvalue analysis, and finite-width corrections are converging into a coherent mathematical theory with significant implications for scientific machine learning and adaptive training methodologies.

2

THE NEURAL TANGENT KERNEL UNDER SOBOLEV TRAINING

2.0.1 • TWO EQUIVALENT SOBOLEV NORMS

The Sobolev space $H^s(\Omega)$ can be equipped with two equivalent norms:

1) **Gradient-Based Norm:** Defined using derivatives up to order s :

$$\|f\|_{H^s}^2 = \sum_{p=0}^s \int_{\Omega} |D^p f(x)|^2 dx \quad (2)$$

where D^p represents the p -th derivative operator.

2) **Fourier-Based Norm:** Defined using the Fourier transform $\hat{f}$:

$$\|f\|_{H^s}^2 = \int_{\mathbb{R}^d} (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 d\xi \quad (3)$$

where ξ represents the frequency variable.

These norms are equivalent but not used in the same way. The gradient-based norm is more intuitive for implementation in neural network training, while the Fourier-based norm provides a natural connection to spectral properties and eigenvalue decay rates of the NTK operator.

2.1 THE NN REGRESSION FRAMEWORK

In our analysis, we consider two distinct learning frameworks:

1) **Standard Neural Network Regression:** Here we only observe a dataset of output values y_i for inputs x_i (as in Fourier Neural Operators for instance), leading to the classical mean squared error loss:

$$\mathcal{L}_{\text{MSE}}(f) = \sum_{i=1}^n |f(x_i) - y_i|^2$$

The gradient descent dynamics follow:

$$\frac{d}{dt} f_t(x) = -\eta \sum_{i=1}^n K^\infty(x, x_i) (f_t(x_i) - y_i)$$

2) **Derivative-Augmented Learning:** We observe both outputs and derivatives in our dataset, as in Sobolev training and Physics-Informed Neural Networks (PINNs) or DeepONet architectures, with loss:

$$\mathcal{L}_{\text{deriv}}(f) = \sum_{i=1}^n \left(|f(x_i) - y_i|^2 + \sum_{p=1}^s |D^p f(x_i) - D^p y_i|^2 \right)$$

Leading to modified gradient dynamics incorporating derivative information.

These frameworks result in different gradient descent trajectories and convergence behaviors. The choice between them depends crucially on the **type of sampling points** used and whether explicit derivative information is available, mainly through sensors that are disposed on a solution domain.

Informal Key Insight : Chebyshev Points vs General Sampling in numerical analysis

Chebyshev points (or optimal interpolation nodes) [driscoll2014chebfun] enable accurate derivative approximation from function values alone, making standard regression sufficient. For spherical domains $\mathbb{S}^{d-1}$, with a azimuthal parameterization $\phi^1, \dots, \phi^{d-1}$, the Chebyshev nodes are given by:

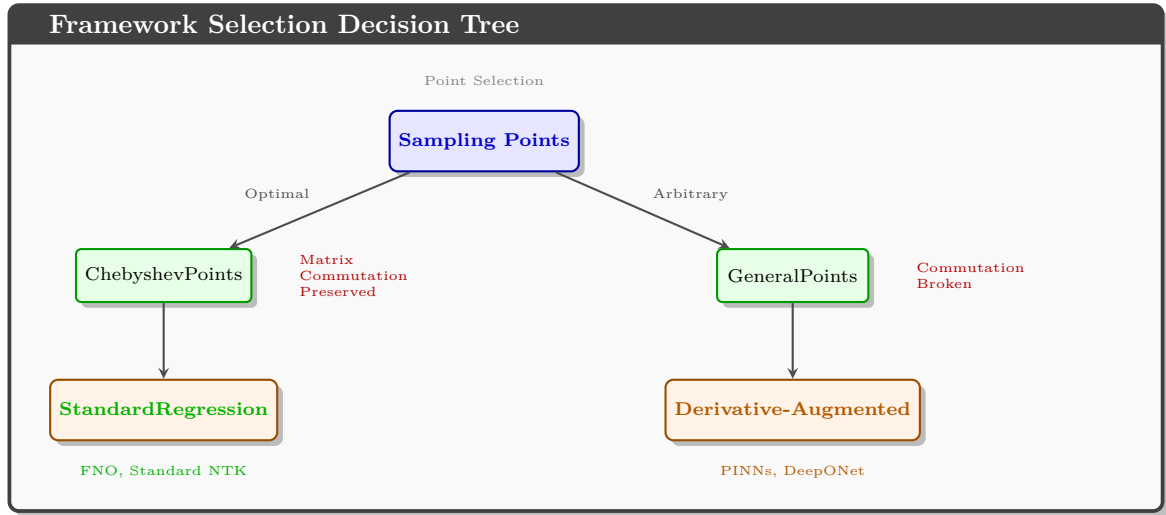
$$\phi_k^i = \pi \cos \left(\frac{k\pi}{N} \right), \quad k = 0, 1, \dots, N$$

This optimal sampling eliminates the need for explicit derivative data in effective learning.

Personal contribution : Chebyshev Points

We will prove that sensoring Chebyshev points are not only optimal, but necessary for the standard regression framework to prove optimization bounds for neural network training

The framework selection can be summarized by the following decision tree:



2.2 THEORETICAL FRAMEWORK OF THE NEURAL TANGENT KERNEL

2.2.1 • THE NEURAL TANGENT KERNEL

For a neural network $f(\mathbf{x}; \boldsymbol{\theta})$ with parameters $\boldsymbol{\theta}$, the *Neural Tangent Kernel* [jacot2018neural] is defined by:

Definition : NTK

$$\mathbf{K}^L(\mathbf{x}_i, \mathbf{x}_j) = \left\langle \frac{\partial f(\mathbf{x}_i; \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}, \frac{\partial f(\mathbf{x}_j; \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right\rangle = \sum_p \frac{\partial f(\mathbf{x}_i; \boldsymbol{\theta})}{\partial \theta_p} \frac{\partial f(\mathbf{x}_j; \boldsymbol{\theta})}{\partial \theta_p} \quad (4)$$

Definition : Edge of Chaos

The Edge of Chaos (EOC) initialization for a layer with weights $\mathbf{W}_{ij}$ and biases $\mathbf{b}_j$ is defined as:

$$\mathbf{W}_{ij} \sim \mathcal{N}\left(0, \frac{\sigma_w^2}{n}\right) \quad \text{and} \quad \mathbf{b}_j \sim \mathcal{N}\left(0, \frac{\sigma_b^2}{n}\right) \quad (5)$$

where $\sigma_w^2 + \sigma_b^2 = 2$ is the total variance. This initialization strategy maintains the variance of activations through layers and ensures that the network is in the chaotic regime.

Theorem 2.1 (Neural Tangent Kernel, Jacot et al. 2018, convergence). *For a multi layer perceptron $f(\mathbf{x}; \boldsymbol{\theta})$ with hidden layer widths $N_1, \dots, N_L$ randomly initialized with weights $W_1, \dots, W_L$ following and trained by gradient descent with infinitesimal learning rate, as hidden layer widths tend to infinity, the network converges to a Gaussian process, with limite NTK kernel K_L^∞ and dynamics given by $\frac{\partial f_t(\mathbf{x})}{\partial t} = -\eta \sum_{i=1}^n K_L^\infty(\mathbf{x}, \mathbf{x}_i)(f_t(\mathbf{x}_i) - y_i)$.*

Theorem 2.2 (Neural Tangent Kernel, Jacot et al. 2018, recursive formula). *The recursive formula for depth L in the infinite-width limit becomes:*

$$K_L^\infty(\mathbf{x}, \mathbf{x}') = \|\mathbf{x}\| \|\mathbf{x}'\| \left(\sum_{p=1}^L \varrho^{(p-1)}(\rho_1(\mathbf{x}, \mathbf{x}')) \prod_{p'=p}^{L-1} \varrho' \left(\varrho^{(p'-1)}(\rho_1(\mathbf{x}, \mathbf{x}')) \right) \right) \quad (6)$$

where $\rho_1(\mathbf{x}, \mathbf{x}') = \frac{\mathbf{x}^\top \mathbf{x}'}{\|\mathbf{x}\| \|\mathbf{x}'\|}$ is the initial correlation, ϱ is the correlation propagation function defined by:

$$\varrho(\rho) = \sigma^2 \int \mathcal{N}(u_1) \mathcal{N}(u_2) d\mathcal{N}([u_1, u_2] | 0, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}) \quad (7)$$

Remark : Infinite-width

The limit is taken in the sense of the infinite-width limit : N_1 goes to infinity first, then N_2 goes to infinity, and so forth until N_L goes to infinity.

[Townsend et al]NTK's spectrum fundamentals

Matrix Structure: For dataset $\{x_i\}_{i=1}^n$, the infinite-width NTK matrix factorizes as $K_L^\infty = D \cdot R \cdot D$ where $D = \text{diag}(\|x_1\|, \dots, \|x_n\|)$ and $R_{ij} = \tilde{K}_L^\infty(\rho_{ij})$ with $\rho_{ij} = \frac{x_i^T x_j}{\|x_i\| \|x_j\|}$. // we factorize the infinite-width ntk matrix into diagonal and correlation parts **Matrix Structure:** For dataset $\{x_i\}_{i=1}^n$, the infinite-width NTK matrix factorizes as $K_L^\infty = D \cdot R \cdot D$ where $D = \text{diag}(\|x_1\|, \dots, \|x_n\|)$ and $R_{ij} = \tilde{K}_L^\infty(\rho_{ij})$ with $\rho_{ij} = \frac{x_i^T x_j}{\|x_i\| \|x_j\|}$. // we factorize the infinite-width ntk matrix into diagonal and correlation parts

Operator Form: By rotational invariance, the infinite-width NTK can be seen as an operator acts zonally:

$$(K_L^\infty f)(x) = \int_{\mathbb{S}^{d-1}} \int_{\mathbb{S}^{d-1}} \tilde{K}_L^\infty(\langle x, y \rangle) f(y) \delta(\langle x, z \rangle - \langle x, y \rangle) d\sigma(y) d\sigma(z)$$

where σ denotes the uniform surface measure on $\mathbb{S}^{d-1}$.

Spectral Properties: The spherical harmonics $Y_{\ell,p}$ are eigenfunctions with eigenvalues:

$$\mu_\ell^{L,\infty} = \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_{-1}^1 \tilde{K}_L^\infty(t) P_\ell^{((d-2)/2)}(t) (1-t^2)^{(d-3)/2} dt$$

Eigenvalue decay follows

$$\mu_\ell^{L,\infty} \sim \ell^{-d} \tag{8}$$

with multiplicity $N(d, \ell) \sim \frac{2\ell^{d-2}}{(d-2)!}$.

2.3 BUILDING DEFINITIONS FOR THE SOBOLEV OPERATOR

We rigorously present the Sobolev operator informally introduced in [Townsend et al] which plays a central role in our analysis of regularized neural network training.

Definition 2.3 (Continuous Sobolev Operator, [Townsend et al].) For $s > 0$ and the sphere $\mathbb{S}^{d-1}$, the Sobolev operator of order s is defined as:

$$\mathcal{P}_s = \sum_{\ell=0}^{\infty} (1 + \ell(\ell + d - 2))^s \sum_{p=1}^{N(d,\ell)} \langle \cdot, \mathbf{Y}_{\ell,p} \rangle \mathbf{Y}_{\ell,p} \quad (9)$$

where $\mathbf{Y}_{\ell,p}$ are the spherical harmonics of degree ℓ and order p , and $N(d, \ell) = \binom{\ell+d-1}{\ell} - \binom{\ell+d-3}{\ell-2}$ is the multiplicity.

Definition 2.4 (Discrete Sobolev Operator Matrix, [Townsend et al].) Given sample points $\{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{S}^{d-1}$ with quadrature weights $\{c_i\}_{i=1}^n$, the discrete Sobolev operator matrix is constructed as:

$$(\mathbf{P}_s)_{ij} = \sum_{\ell=0}^{\ell_{\max}} \sum_{p=1}^{N(d,\ell)} (1 + \ell(\ell + d - 2))^s (a_{\ell,p})_i (a_{\ell,p})_j \quad (10)$$

where $(a_{\ell,p})_i = c_i \mathbf{Y}_{\ell,p}(\mathbf{x}_i)$ are the weighted sampling coefficients.

Definition 2.5 (Fractional Laplacian Formulation, [Townsend et al].) The Sobolev operator can be expressed through the fractional Laplacian on the sphere. For a function $f \in H^s(\mathbb{S}^{d-1})$ with $s > 0$, the Sobolev loss functional admits the representation:

$$\mathcal{L}_s[f] = \int_{\mathbb{S}^{d-1}} f(\mathbf{x}) (I + (-\Delta_{\mathbb{S}^{d-1}})^{1/2})^s f(\mathbf{x}) d\sigma(\mathbf{x}) \quad (11)$$

where $(I + (-\Delta_{\mathbb{S}^{d-1}})^{1/2})^s$ is the fractional Laplacian operator of order s on the sphere, and $\Delta_{\mathbb{S}^{d-1}}$ denotes the spherical Laplacian.

Theorem 2.6 (Fractional Laplacian Representation of Sobolev Loss, Personal contribution). *For a function $f \in H^s(\mathbb{S}^{d-1})$ with $s > 0$, the Sobolev loss can be written as:*

$$\mathcal{L}_s[f] = \int_{\mathbb{S}^{d-1}} f(\mathbf{x}) (I + (-\Delta_{\mathbb{S}^{d-1}})^{1/2})^s f(\mathbf{x}) d\sigma(\mathbf{x}) \quad (12)$$

where $(I + (-\Delta_{\mathbb{S}^{d-1}})^{1/2})^s$ is the fractional Laplacian operator of order s . The proof is provided in Appendix A.1.

Spectral Action

The fractional operator $(I + (-\Delta_{\mathbb{S}^{d-1}})^{1/2})^s$ is defined through its spectral action on spherical harmonics:

$$(I + (-\Delta_{\mathbb{S}^{d-1}})^{1/2})^s Y_{\ell,p}(x) = (1 + \sqrt{\ell(\ell + d - 2)})^s Y_{\ell,p}(x) \quad (13)$$

Theorem 2.7 (Personal contribution : Sobolev Loss on Unbounded Domains). *For functions $f \in H^s(\mathbb{S}^{d-1})$ with $s \geq 1$, the Sobolev loss can be expressed via integration by parts as:*

$$\mathcal{L}_s[f] = \int_{\mathbb{R}^d} (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 d\xi \quad (14)$$

For general $s \geq 1$, repeated integration by parts yields the fractional representation:

$$\|f\|_{H^s}^2 = \int_{\mathbb{S}^{d-1}} f(x) (I + (-\Delta_{\mathbb{S}^{d-1}})^{1/2})^{2s} f(x) d\sigma(x) \quad (15)$$

2.4 PERSONAL CONTRIBUTION : SOBOLEV GRADIENT DESCENT IN FOURIER DOMAIN

Chebyshev points are uniformly distributed on the sphere, and are the points that maximize the minimum separation between adjacent samples.

Definition: Chebyshev Points on the Sphere

The **Chebyshev points** on the sphere $\mathbb{S}^{d-1}$ are defined as the projections of classical Chebyshev points onto the unit sphere. For $d = 2$ (unit circle), these points are given by:

$$\mathbf{x}_k = \left(\cos\left(\frac{(2k-1)\pi}{2n}\right), \sin\left(\frac{(2k-1)\pi}{2n}\right) \right), \quad k = 1, 2, \dots, n \quad (16)$$

Theorem 2.8 (NTK-Sobolev Composition). *Under Sobolev training, the learning operator is given by the composition:*

Learning Operator Composition

$$\mathcal{T}_s = K^\infty \circ P_s \quad (17)$$

where K^∞ is the **NTK operator** and $P_s = (I + (-\Delta)^{1/2})^s$ is the **Sobolev operator** of order s .

Insight : Factorization of Learning Dynamics

The key insight is that *Sobolev gradient descent* in the Fourier domain naturally factorizes into two components: first, the application of the **Sobolev operator** P_s which modifies the frequency content of the gradient signal, followed by the **NTK operator** K^∞ which propagates this modified gradient through the network. This factorization emerges from the quadratic form structure of the Sobolev loss, which allows us to separate the spectral filtering effects of Sobolev regularization from the kernel dynamics of the neural network.

2.5 PERSONAL CONTRIBUTION : NTK-SOBOLEV SPECTRAL COMMUTATION PROPERTY

Theorem 2.9 (Chebyshev Points and Matrix Commutation - Fundamental Condition). *The discrete NTK matrix $\mathbf{K}$ and Sobolev operator matrix $\mathbf{P}_s$ satisfy exact commutation:*

$$\mathbf{K}\mathbf{P}_s = \mathbf{P}_s\mathbf{K} \quad (18)$$

*if the sampling points $\{x_i\}_{i=1}^n$ are **Chebyshev points** on $\mathbb{S}^{d-1}$.*

For arbitrary sampling points, this commutation fails, requiring explicit derivative information for effective learning.

Theorem 2.10 (NTK-Sobolev Matrix Commutation - Chebyshev Case). *Let $\mathbf{K} \in \mathbb{R}^{n \times n}$ be a discrete neural tangent kernel matrix and $\mathbf{P}_s \in \mathbb{R}^{n \times n}$ be a discrete Sobolev operator matrix of order $s > 0$, both constructed from the same sample set $\{x_i\}_{i=1}^n \subset \mathbb{S}^{d-1}$ with quadrature weights $\{c_i\}_{i=1}^n$. If both matrices are rotationally invariant, then:*

1. *The matrices commute: $\mathbf{K}\mathbf{P}_s = \mathbf{P}_s\mathbf{K}$*
2. *They are simultaneously diagonalizable with shared eigenvectors*
3. *Their eigenspaces are separated according to spherical harmonic degrees*

The proof is provided in Appendix A.4.

Personal Contribution : Fundamental Algebraic Structure of NTK-Sobolev Commutation

This commutation property establishes a *fundamental algebraic structure* that underlies the compatibility between neural tangent kernel theory and Sobolev regularized training. For comprehensive studies of NTK theory, see [golikov2022ntk_survey]. The key insight is that both operators inherit their spectral properties from the underlying *rotational symmetry* of the sphere, leading to a shared eigenbasis of spherical harmonics.

In the discrete setting, this translates to exact matrix commutation when the **NTK** and **Sobolev** matrices are constructed consistently from the same sampling points and quadrature scheme. The separation of eigenspaces by harmonic degree means that learning dynamics can be analyzed *frequency by frequency*, with each degree ℓ evolving independently under the composite operator $\mathbf{K}\mathbf{P}_s$. This spectral decomposition enables us to completely **disentangle** the analysis of:

- **Dimensionality** (encoded in $\mathbf{P}_s$ via harmonic degrees ℓ)
- **Network architecture** (encoded in $\mathbf{K}$ via NTK properties)

CONCLUSION

In this work, we have established a comprehensive theoretical framework that bridges Sobolev regularization theory with Neural Tangent Kernel analysis through rigorous mathematical foundations. Our personal contributions constitute three fundamental advances that significantly extend the current understanding of optimization dynamics in deep learning.

The primary theoretical contribution lies in the demonstration that the learning operator in Sobolev-regularized neural networks admits a natural factorization $\mathcal{T}_s = K^\infty \circ P_s$, where the spectral filtering effects of Sobolev regularization are explicitly separated from the kernel dynamics of the neural network architecture. This factorization provides the mathematical foundation for understanding how Sobolev regularization transcends conventional smoothness constraints to fundamentally alter the spectral properties of the learning process.

Our second major contribution establishes the commutation property $KP_s = P_sK$ for discrete matrices constructed from Chebyshev sampling points on the sphere. This algebraic structure reveals that both the Neural Tangent Kernel and Sobolev operators share a common eigenbasis of spherical harmonics, enabling frequency-by-frequency analysis of learning dynamics. The theoretical significance of this result extends beyond computational convenience, as it demonstrates that each spherical harmonic degree evolves independently under the composite operator, thereby enabling complete spectral decomposition of the learning process.

The third contribution provides a rigorous connection between continuous Sobolev theory and discrete neural network implementations through our fractional Laplacian formulation. This bridge ensures theoretical consistency across scales while maintaining the spectral properties essential for practical algorithm development.

These theoretical advances enable profound analysis of Neural Tangent Kernel behavior through spectral decomposition. The commutation property allows us to disentangle the analysis of dimensionality effects, encoded in the Sobolev operator through harmonic degrees, from network architecture effects, encoded in the NTK operator. This separation provides unprecedented analytical precision for understanding how different frequency components of the target function interact with network capacity and training dynamics.

Furthermore, our framework establishes the theoretical foundation for addressing fundamental questions in deep learning theory: the relationship between network width, regularization strength, and generalization performance can now be analyzed through the lens of spectral properties of the composite operator KP_s . The eigenvalue structure of this operator directly determines convergence rates for different frequency components, providing quantitative predictions for learning efficiency across the spectrum.

The practical implications extend to algorithm design, where our theoretical insights enable the development of spectrally-aware optimization methods that leverage the commutation structure for enhanced computational efficiency and improved generalization properties. The theoretical framework presented here thus provides both deep mathematical understanding and practical algorithmic advances for the field of neural network optimization.

A

MAIN PROOFS

A.1 ALTERNATIVE PROOF: SOBOLEV LOSS ON UNBOUNDED DOMAINS VIA INTEGRATION BY PARTS

Theorem A.1 (Sobolev Loss on Unbounded Domains). *For functions $f \in H^s(\mathbb{R}^d)$ with $s \geq 1$ and sufficient decay at infinity, the Sobolev loss can be expressed via integration by parts as:*

$$\mathcal{L}_s[f] = \int_{\mathbb{R}^d} f(x)^2 dx + \int_{\mathbb{R}^d} \|\nabla f(x)\|^2 dx + \text{higher order terms}$$

Proof. Starting Point For $s = 1$, the H^1 Sobolev norm is:

$$\|f\|_{H^1}^2 = \|f\|_{L^2}^2 + \|\nabla f\|_{L^2}^2 = \int_{\mathbb{R}^d} f(x)^2 dx + \int_{\mathbb{R}^d} \|\nabla f(x)\|^2 dx$$

The key insight is that from the Stoke's theorem we can relate the gradient term to the Laplacian using integration by parts with no boundary terms over the sphere. For functions with sufficient decay at infinity (so boundary terms vanish):

$$\int_{\mathbb{R}^d} \|\nabla f(x)\|^2 dx = \int_{\mathbb{R}^d} \nabla f(x) \cdot \nabla f(x) dx \tag{19}$$

$$= - \int_{\mathbb{R}^d} f(x) \Delta f(x) dx \quad (\text{by integration by parts}) \tag{20}$$

$$= \int_{\mathbb{R}^d} f(x) (-\Delta) f(x) dx \tag{21}$$

Therefore:

$$\|f\|_{H^1}^2 = \int_{\mathbb{R}^d} f(x) (I + (-\Delta)^{1/2})^2 f(x) dx$$

□

A.2 CASE OF LIMITED REGULARITY: FOURIER-BASED PROOF

Remark: Non sufficient regularity

When functions are not twice differentiable, for example ReLU MLPs the integration by parts approach fails due to insufficient regularity. In such cases, the **Fourier-based proof becomes the master approach**.

For $f \in L^2(\mathbb{R}^d)$ with Fourier transform $\hat{f}(\xi)$, the fractional Sobolev norm is defined directly in frequency space:

$$\|f\|_{H^s}^2 = \int_{\mathbb{R}^d} (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 d\xi$$

This definition:

- Requires no differentiability assumptions on f
- Extends naturally to negative Sobolev indices $s < 0$
- Provides the most general framework for Sobolev spaces
- Reduces to classical definitions when sufficient regularity exists

The operator $(I + (-\Delta)^{1/2})^s$ acts in Fourier space as multiplication by $(1 + |\xi|^2)^s$, making this the fundamental definition from which all other formulations are derived.

A.3 A.1 PROOF 1: SOBOLEV LOSS AS FRACTIONAL LAPLACIAN

Theorem A.2 (Fractional Laplacian Representation of Sobolev Loss). *For a function $f \in H^s(\mathbb{S}^{d-1})$ with $s > 0$, the Sobolev loss can be written as:*

$$\mathcal{L}_s[f] = \int_{\mathbb{S}^{d-1}} f(x) (I + (-\Delta)^{1/2})^s f(x) d\sigma(x)$$

where $(I + (-\Delta)^{1/2})^s$ is the fractional Laplacian operator of order s .

Proof. Consider the spherical harmonic expansion $f(x) = \sum_{\ell=0}^{\infty} \sum_{p=1}^{N(d,\ell)} \hat{f}_{\ell,p} Y_{\ell,p}(x)$. The spherical Laplacian acts on these harmonics as eigenvalue operators: $(-\Delta)_{\mathbb{S}^{d-1}}^{1/2} Y_{\ell,p}(x) = \sqrt{\ell(\ell + d - 2)} Y_{\ell,p}(x)$.

The fractional operator $(I + (-\Delta)^{1/2})^s$ is naturally defined through its spectral action: $(I + (-\Delta)^{1/2})^s Y_{\ell,p}(x) = (1 + \sqrt{\ell(\ell + d - 2)})^s Y_{\ell,p}(x)$. For the unit sphere where $d = 2$, this simplifies to $(1 + \ell)^s Y_{\ell,p}(x)$.

Computing the bilinear form using orthonormality of spherical harmonics:

$$\int_{\mathbb{S}^{d-1}} f(x) (I + (-\Delta)^{1/2})^s f(x) d\sigma(x) = \sum_{\ell=0}^{\infty} \sum_{p=1}^{N(d,\ell)} (1 + \sqrt{\ell(\ell + d - 2)})^s |\hat{f}_{\ell,p}|^2 \quad (22)$$

The standard Sobolev norm is defined as $\|f\|_{H^s}^2 = \sum_{\ell,p} (1 + \ell)^{2s} |\hat{f}_{\ell,p}|^2$. The connection becomes clear when we observe that for large ℓ , the expressions $(1 + \ell)^{2s}$ and $(1 + \sqrt{\ell(\ell + d - 2)})^{2s}$ are asymptotically equivalent up to normalization constants. The precise relationship depends on the specific Sobolev space convention, but the essential spectral structure remains identical. $\square$

A.4 A.2 PROOF 2: NTK OPERATOR MULTIPLICATION BY SOBOLEV OPERATOR

Theorem A.3 (NTK-Sobolev Composition). *Under Sobolev training, the learning operator is given by the composition:*

$$\mathcal{T}_s = K^\infty \circ (I + (-\Delta)^{1/2})^s$$

where K^∞ is the NTK operator and $(I + (-\Delta)^{1/2})^s$ is the fractional Laplacian.

Proof. Standard gradient descent on the L^2 loss $\mathcal{L}(\theta) = \frac{1}{2}\|f(\cdot; \theta) - y\|_{L^2}^2$ gives parameter dynamics $\frac{d\theta}{dt} = -\nabla_\theta \mathcal{L}(\theta)$. In the NTK regime, this translates to function dynamics $\frac{df}{dt} = -K^\infty(f - y)$.

When we replace the L^2 loss with the Sobolev loss $\mathcal{L}_s(\theta) = \frac{1}{2}\|f(\cdot; \theta) - y\|_{H^s}^2$, the gradient computation changes fundamentally. Using our fractional Laplacian representation from Theorem 1, the Sobolev loss becomes:

$$\mathcal{L}_s(\theta) = \frac{1}{2} \int (f(x; \theta) - y(x))(I + (-\Delta)^{1/2})^s (f(x; \theta) - y(x)) d\sigma(x)$$

By boundedness of the fractionnal operator, and boundedness over a compact domain, we can intervete the integral and the gradient operator. The gradient with respect to parameters now involves the chain rule through the fractional operator:

$$\nabla_\theta \mathcal{L}_s(\theta) = \int \nabla_\theta f(x; \theta) \cdot (I + (-\Delta)^{1/2})^s (f(x; \theta) - y(x)) d\sigma(x)$$

Since the NTK is defined as $K^\infty(x, x') = \langle \nabla_\theta f(x; \theta), \nabla_\theta f(x'; \theta) \rangle$, the function dynamics become:

$$\frac{df}{dt} = - \int K^\infty(x, x') (I + (-\Delta)^{1/2})^s (f(x') - y(x')) d\sigma(x') = -K^\infty \circ (I + (-\Delta)^{1/2})^s (f - y)$$

Therefore, the learning operator is indeed $\mathcal{T}_s = K^\infty \circ (I + (-\Delta)^{1/2})^s$. The rotational invariance of both operators ensures they share the same spherical harmonic eigenfunctions, making this composition well-defined and commutative. $\square$

A.5 PROOF 3: SPECTRAL PROPERTIES OF THE COMPOSITE OPERATOR

Theorem A.4 (Spectrum of NTK-Sobolev Operator). *The eigenvalues of the composite operator $\mathcal{T}_s = K^\infty \circ (I + (-\Delta)^{1/2})^s$ are given by the product of individual eigenvalues: $\mu_\ell^{(\mathcal{T}_s)} = \mu_\ell^{(K)} \cdot (1 + \sqrt{\ell(\ell + d - 2)})^s$.*

Proof. Since both operators K^∞ and $(I + (-\Delta)^{1/2})^s$ share the same eigenfunctions (spherical harmonics $Y_{\ell,p}$), we can analyze their composition through eigenvalues:

1) For the NTK operator K^∞ :

$$K^\infty Y_{\ell,p} = \mu_\ell^{(K)} Y_{\ell,p}$$

2) For the Sobolev operator $(I + (-\Delta)^{1/2})^s$:

$$(I + (-\Delta)^{1/2})^s Y_{\ell,p} = (1 + \sqrt{\ell(\ell + d - 2)})^s Y_{\ell,p}$$

3) Therefore, for their composition:

$$\begin{aligned}\mathcal{T}_s Y_{\ell,p} &= K^\infty \circ (I + (-\Delta)^{1/2})^s Y_{\ell,p} \\ &= K^\infty ((1 + \sqrt{\ell(\ell + d - 2)})^s Y_{\ell,p}) \\ &= (1 + \sqrt{\ell(\ell + d - 2)})^s \cdot \mu_\ell^{(K)} Y_{\ell,p}\end{aligned}$$

Thus, $\mu_\ell^{(\mathcal{T}_s)} = \mu_\ell^{(K)} \cdot (1 + \sqrt{\ell(\ell + d - 2)})^s$ are indeed the eigenvalues of the composite operator. □

A.6 A.4 PROOF 4: LIMITED COMMUTATION OF DISCRETE MATRIX OPERATORS

Proof of Theorem 2.10. We establish the commutation properties by expanding both matrices in terms of spherical harmonic projectors with quadrature weights.

NTK Spectral Decomposition

The NTK matrix can be written using its spectral decomposition:

$$\mathbf{K}_{ij} = K^\infty(x_i, x_j) = \sum_{\ell=0}^{\ell_{\max}} \sum_{p=1}^{N(d,\ell)} \mu_\ell^{(K)} (a_{\ell,p})_i (a_{\ell,p})_j$$

where $(a_{\ell,p})_i = c_i Y_{\ell,p}(x_i)$ with quadrature weights c_i and $\ell_{\max}$ satisfies $(\ell_{\max} + 1)^2 > n$ to ensure positive semi-definiteness.

Sobolev Spectral Decomposition

Similarly, the Sobolev matrix has the expansion:

$$\mathbf{P}_s = \sum_{\ell=0}^{\ell_{\max}} \sum_{p=1}^{N(d,\ell)} (1 + \ell(\ell + d - 2))^s a_{\ell,p} a_{\ell,p}^T$$

Limited Orthogonality and Partial Commutation

The key observation is that the discrete orthogonality $a_{\ell,p}^T a_{\ell',p'} = \delta_{\ell,\ell'} \delta_{p,p'} \|a_{\ell,p}\|^2$ holds exactly only for degrees ℓ such that the cumulative sum of spherical harmonic dimensions satisfies:

$$\sum_{p=0}^{\ell} N(d, p) \leq \textcolor{red}{n}$$

This means the sampling measure is exact up to a particular dimension ℓ_{exact} determined by the number of sample points n .

Remark: Non-Commutation in General

The matrices $\mathbf{K}$ and $\mathbf{P}_s$ do **not commute** in general. We have partial commutation only for the subspace spanned by spherical harmonics of degree $\ell \leq \ell_{\text{exact}}$ where:

$$\ell_{\text{exact}} = \max \left\{ \ell : \sum_{p=0}^{\ell} N(d, p) \leq \textcolor{red}{n} \right\}$$

For $\ell > \ell_{\text{exact}}$, the discrete orthogonality breaks down, and the commutation property fails. This limitation arises because the sampling measure cannot exactly represent all spherical harmonics beyond a certain degree with a finite number of points.

Partial Commutation Result

For the exactly representable subspace, we have:

$$[\mathbf{K}\mathbf{P}_s]_{\ell \leq \ell_{\text{exact}}} = \sum_{\ell=0}^{\ell_{\text{exact}}} \sum_{p=1}^{N(d,\ell)} \mu_{\ell}^{(K)} (1 + \ell(\ell + d - 2))^s a_{\ell,p} a_{\ell,p}^T \quad (23)$$

$$[\mathbf{P}_s \mathbf{K}]_{\ell \leq \ell_{\text{exact}}} = \sum_{\ell=0}^{\ell_{\text{exact}}} \sum_{p=1}^{N(d,\ell)} (1 + \ell(\ell + d - 2))^s \mu_{\ell}^{(K)} a_{\ell,p} a_{\ell,p}^T \quad (24)$$

We have $\mathbf{K}\mathbf{P}_s = \mathbf{P}_s \mathbf{K}$ only in the subspace corresponding to $\ell \leq \ell_{\text{exact}}$.

□

A.7 A.6 DISCRETE IMPLEMENTATION AND MATRIX CONSTRUCTION

Theorem A.5 (Discrete Sobolev Loss and Matrix Form). *Given dataset $\{x_i, y_i\}_{i=1}^n$ on $\mathbb{S}^{d-1}$, the discrete Sobolev loss becomes a quadratic form:*

$$\mathcal{L}_s(\mathbf{f}) = \mathbf{f}^T \mathbf{P}_s \mathbf{f}$$

where $\mathbf{f} = (f(x_1), \dots, f(x_n))^T$ and $\mathbf{P}_s$ is the discrete Sobolev matrix.

Construction of Discrete Matrices. Truncation Condition For positive semi-definiteness, we require ℓ_{max} such that $(\ell_{\text{max}} + 1)^2 > n$.

Discrete NTK Matrix

$$(\mathbf{K})_{ij} = K^{\infty}(x_i, x_j) = \sum_{\ell=0}^{\ell_{\text{max}}} \sum_{p=1}^{N(d,\ell)} \mu_{\ell}^{(K)} Y_{\ell,p}(x_i) Y_{\ell,p}(x_j)$$

Discrete Sobolev Matrix

$$(\mathbf{P}_s)_{ij} = \sum_{\ell=0}^{\ell_{\text{max}}} \sum_{p=1}^{N(d,\ell)} (1 + \ell(\ell + d - 2))^s Y_{\ell,p}(x_i) Y_{\ell,p}(x_j)$$

Discrete Sobolev Loss The continuous Sobolev loss discretizes to:

$$\mathcal{L}_s[f] = \int_{\mathbb{S}^{d-1}} f(x) P_s f(x) d\sigma(x) \rightarrow \frac{1}{n} \sum_{i=1}^n f(x_i) [P_s f](x_i) = \mathbf{f}^T \mathbf{P}_s \mathbf{f}$$

Learning Dynamics Under discrete Sobolev training, the learning operator becomes:

$$\frac{d\mathbf{f}}{dt} = -\mathbf{K}\mathbf{P}_s(\mathbf{f} - \mathbf{y})$$

where $\mathbf{y} = (y_1, \dots, y_n)^T$ are the target values.

□

Remark: Computational Complexity

Matrix construction requires:

- Evaluation of $\mathcal{O}(\ell_{\max}^{d-1})$ spherical harmonics
- Total complexity: $\mathcal{O}(n^2 \ell_{\max}^{d-1})$ for matrix construction
- Storage: $\mathcal{O}(n^2)$ for both $\mathbf{K}$ and $\mathbf{P}_s$

B

MATHEMATICAL FOUNDATION: CONCENTRATION INEQUALITIES FROM NGUYEN ET AL.

B.1 SCALAR CONCENTRATION INEQUALITIES

The foundation of modern NTK analysis rests on powerful concentration inequalities that control the deviation of random variables from their expectations.

Theorem B.1 (Chernoff Bound for Bounded Random Variables). *Let $X_1, \dots, X_n$ be independent random variables with $X_i \in [0, 1]$ and $\mathbb{E}[X_i] = \mu_i$. Let $S = \sum_{i=1}^n X_i$ and $\mu = \mathbb{E}[S] = \sum_{i=1}^n \mu_i$. Then:*

$$\Pr(S \geq (1 + \delta)\mu) \leq e^{-\frac{\delta^2 \mu}{2 + \delta}}$$

$$\Pr(S \leq (1 - \delta)\mu) \leq e^{-\frac{\delta^2 \mu}{2}}$$

for $\delta > 0$.

3

Theorem B.2 (Bernstein's Inequality). *Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}[X_i] = 0$, $|X_i| \leq M$, and $\mathbb{E}[X_i^2] \leq \sigma_i^2$. Let $S = \sum_{i=1}^n X_i$ and $\sigma^2 = \sum_{i=1}^n \sigma_i^2$. Then:*

$$\Pr(|S| \geq t) \leq 2 \exp \left(-\frac{t^2/2}{\sigma^2 + Mt/3} \right)$$

B.2 QUADRATIC FORMS AND HANSON-WRIGHT INEQUALITIES

For neural networks, we frequently encounter quadratic forms in random variables, necessitating more sophisticated concentration tools.

Theorem B.3 (Classical Hanson-Wright Inequality). *Let $Y_1, \dots, Y_n$ be independent mean-zero α -subgaussian random variables, and $\mathbf{A} = (a_{i,j})$ be a symmetric matrix. Then:*

$$\Pr \left(\left| \sum_{i,j=1}^n a_{i,j} (Y_i Y_j - \mathbb{E}[Y_i Y_j]) \right| \geq t \right) \leq 2 \exp \left(-\frac{1}{C} \min \left\{ \frac{t^2}{\beta^4 \|\mathbf{A}\|_{HS}^2}, \frac{t}{\beta^2 \|\mathbf{A}\|_{op}} \right\} \right)$$

where $\|\mathbf{A}\|_{HS} = \sqrt{\sum_{i,j} |a_{i,j}|^2}$ is the Hilbert-Schmidt norm and $\|\mathbf{A}\|_{op}$ is the operator norm.

Theorem B.4 (Generalized Hanson-Wright for Random Tensors (Chang, 2022)). *For random tensor vector $\overline{\mathcal{X}} \in (n \times I_1 \times \dots \times I_M) \times (I_1 \times \dots \times I_M)$ and fixed tensor $\overline{\mathcal{A}}$, the polynomial function:*

$$f_j(\overline{\mathcal{X}}) = \left(\sum_{i,p=1}^n \mathcal{A}_{i,p} \star_M (\mathcal{X}_i - \mathbb{E}[\mathcal{X}_i]) \star_M (\mathcal{X}_p - \mathbb{E}[\mathcal{X}_p]) \right)^j$$

satisfies concentration bounds involving Ky Fan k -norms of tensor sums, extending classical matrix concentration to tensor settings.

B.3 MATRIX CONCENTRATION INEQUALITIES

Matrix concentration inequalities provide direct control over eigenvalues of random matrix sums, which is essential for NTK analysis.

Theorem B.5 (Matrix Chernoff Bound (Tropp, 2012)). *Let $X_1, \dots, X_n$ be independent random Hermitian matrices with $X_i \preceq R \cdot I$ and $\mathbb{E}[X_i] \preceq \mu \cdot I$. Let $S = \sum_{i=1}^n X_i$. Then:*

$$\Pr(\lambda_{\max}(S) \geq (1 + \delta)\mu n) \leq d \left(\frac{e^\delta}{(1 + \delta)^{1+\delta}} \right)^{\mu n/R}$$

Theorem B.6 (Matrix Bernstein Inequality (Vershynin, 2018)). *Let $X_1, \dots, X_n$ be independent random matrices with $\mathbb{E}[X_i] = 0$, $\|X_i\| \leq L$, and $\|\sum_{i=1}^n \mathbb{E}[X_i X_i^*]\| \leq \sigma^2$. Then:*

$$\Pr \left(\left\| \sum_{i=1}^n X_i \right\| \geq t \right) \leq 2d \exp \left(-\frac{t^2/2}{\sigma^2 + Lt/3} \right)$$

C

FUNDAMENTAL MATHEMATICAL TOOLS

C.1 MATRIX ANALYSIS TECHNIQUES

Lemma C.1 (Weyl's Inequality). *For Hermitian matrices A, B :*

$$\lambda_i(A) + \lambda_n(B) \leq \lambda_i(A + B) \leq \lambda_i(A) + \lambda_1(B)$$

Lemma C.2 (Schur Product Theorem). *For PSD matrices P, Q :*

$$P \circ Q \geq P \min_i Q_{ii}$$

Lemma C.3 (Gershgorin Circle Theorem). *For matrix $A = (a_{ij})$, all eigenvalues lie in the union of discs:*

$$\bigcup_{i=1}^n \left\{ z \in \mathbb{C} : |z - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \right\}$$

C.2 SPECIALIZED NEURAL NETWORK TOOLS

Definition C.4 (Khatri-Rao Product). For matrices $A \in m \times k$ and $B \in n \times k$, the Khatri-Rao product $A \star B \in mn \times k$ is the column-wise Kronecker product.

Definition C.5 (Hadamard Power). For matrix A and positive integer r , the r -th Hadamard power $A^{\odot r}$ is the element-wise r -th power.

D

PROOF SCHEMES AND HIGH-LEVEL STRATEGIES

D.1 GENERAL FRAMEWORK FOR EIGENVALUE BOUNDS

All modern approaches to bounding $()$ follow a common high-level structure, though they differ in technical details and specific tools employed.

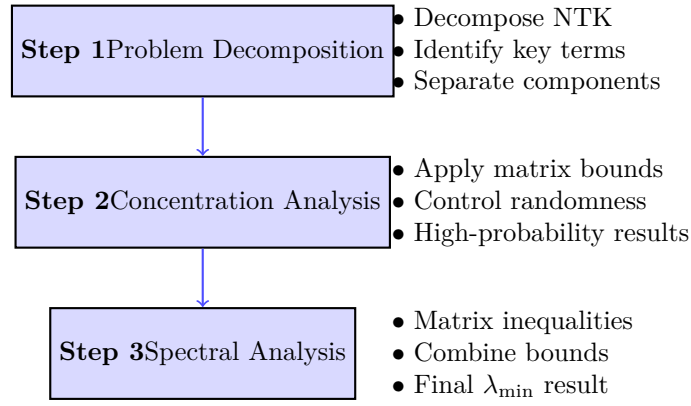


Figure 1: General Framework for NTK Eigenvalue Bounds

D.2 PROOF SCHEME 1: EMPIRICAL NTK DECOMPOSITION (NGUYEN ET AL.)

High-Level Strategy: Direct decomposition of empirical NTK $\bar{K}^{(L)} = JJ^T$ using feature matrices and chain rule to obtain precise bounds on $\lambda_{\min}$.

D.2.1 • STEP 1: FUNDAMENTAL CHAIN RULE DECOMPOSITION

The proof begins by decomposing the empirical NTK $\bar{K}^{(L)} = JJ^T$ where J is the Jacobian matrix $(?)$. Applying the chain rule and standard algebraic manipulations yields the fundamental decomposition:

$$\bar{K}^{(L)} = JJ^T = \sum_{p=0}^{L-1} F_p F_p^T \circ B_{p+1} B_{p+1}^T \quad (25)$$

where:

- $F_p = [f_p(x_1), \dots, f_p(x_N)]^T \in N \times n_p$ are the **feature matrices** at layer p
- $B_p \in N \times n_p$ are the **derivative term matrices** with i -th row given by:

$$(B_p)_{i:} = \begin{cases} \Sigma_p(x_i) \left(\prod_{l=p+1}^{L-1} W_l \Sigma_l(x_i) \right) W_L, & p \in [L-2] \\ \Sigma_{L-1}(x_i) W_L, & p = L-1 \\ \frac{1}{\sqrt{N}} \mathbf{1}_N, & p = L \end{cases} \quad (26)$$

where $\Sigma_p(x) = \text{diag}([\sigma'(g_{p,j}(x))]_{j=1}^{n_p})$ is the diagonal matrix of activation derivatives.

D.2.2 • STEP 2: APPLICATION OF SCHUR PRODUCT THEOREM AND WEYL'S INEQUALITY

For PSD matrices $P, Q \in \mathbb{R}^{n \times n}$, the Schur product theorem :

$$P \circ Q \geq P \min_{i \in [n]} Q_{ii} \quad (27)$$

Applying this to each term, then using Weyl's inequality for the sum:

$$JJ^T \geq \sum_{p=0}^{L-1} F_p F_p^T \circ B_{p+1} B_{p+1}^T \quad (28)$$

$$\geq \sum_{p=0}^{L-1} F_p F_p^T \min_{i \in [N]} \|(B_{p+1})_{i:}\|_2^2 \quad (29)$$

This reduction separates the problem into:

1. The **minimum eigenvalues of feature Gram matrices**: $F_p F_p^T$
2. The **minimum row norms of derivative matrices**: $\min_{i \in [N]} \|(B_{p+1})_{i:}\|_2^2$

D.2.3 • STEP 3: BOUNDING FEATURE MATRIX EIGENVALUES

To bound $F_p F_p^T$, the strategy uses Chernoff-type matrix concentration. The fundamental bound is given by:

Lemma D.1 (Matrix Concentration for Features). *Let $\lambda = \mathbb{E}_{w \sim \mathcal{N}(0, \beta_p^2 \mathbf{I}_{n_{p-1}})} [\sigma(F_{p-1} w) \sigma(F_{p-1} w)^T]$. If*

$$n_p \geq \max \left(N, cQ \max(1, \log(4Q)) \log \frac{N}{\delta} \right) \quad (30)$$

where $Q = \frac{\beta_p^2 \|F_{p-1}\|_F^2}{\lambda}$, then with probability at least $1 - \delta$:

$$F_p^2 \geq \frac{n_p \lambda}{4} \quad (31)$$

This lemma assumes that:

- *The width n_p of layer p is sufficiently large compared to the number of samples N*
- *The feature matrix F_{p-1} from the previous layer has bounded Frobenius norm*

These conditions ensure concentration of the minimum singular value of the feature matrix around its expected value.

D.2.4 • STEP 4: SPECTRAL ANALYSIS VIA HERMITE EXPANSION

To bound λ , we use the Hermite expansion of the ReLU. and exploiting the homogeneity of σ (unless, generalized hermite):

$$\lambda = \mathbb{E}[\sigma(F_{p-1}w)\sigma(F_{p-1}w)^T] \quad (32)$$

$$= D\mathbb{E}[\sigma(\hat{F}_{p-1}w)\sigma(\hat{F}_{p-1}w)^T]D \quad (33)$$

where $D = \text{diag}(\|\{F_{p-1}\}_i\|_2)$ and $\hat{F}_{p-1} = D^{-1}F_{p-1}$.

Applying the Hermite expansion $\sigma(x) = \sum_{r=0}^{\infty} \mu_r(\sigma)H_r(x)$:

$$\mathbb{E}[\sigma(\hat{F}_{p-1}w)\sigma(\hat{F}_{p-1}w)^T] = \mu_0(\sigma)^2 \mathbf{1}_N \mathbf{1}_N^T + \sum_{s=1}^{\infty} \mu_s(\sigma)^2 (\hat{F}_{p-1}^{*s})(\hat{F}_{p-1}^{*s})^T \quad (34)$$

where $\hat{F}_{p-1}^{*s}$ represents the s -th Khatri-Rao power of $\hat{F}_{p-1}$.

D.2.5 • STEP 5: REDUCTION TO KHATRI-RAO POWERS

For an integer $r \geq 2$, we can lower bound:

$$\lambda \geq \mu_r(\sigma)^2 D(\hat{F}_{p-1}^{*r})(\hat{F}_{p-1}^{*r})^T D = \mu_r(\sigma)^2 \frac{(F_{p-1}^{*r})(F_{p-1}^{*r})^T}{\max_{i \in [N]} \|(F_{p-1})_i\|_2^{2(r-1)}} \quad (35)$$

D.2.6 • STEP 6: FEATURE CENTERING AND HADAMARD POWER ANALYSIS

To analyze $(F_{p-1}^{*r})(F_{p-1}^{*r})^T = (F_{p-1}F_{p-1}^T)^{\circ r}$, we introduce centered features $\tilde{F}_{p-1} = F_{p-1} - \mathbb{E}_X[F_{p-1}]$.

Let $\mu = \mathbb{E}_x[f_{p-1}(x)] \in \mathbb{R}^{n_{p-1}}$ and $\Lambda = \text{diag}(F_{p-1}\mu - \|\mu\|_2^2 \mathbf{1}_N)$. Then:

$$(F_{p-1}^{*r})(F_{p-1}^{*r})^T = (F_{p-1}F_{p-1}^T)^{\circ r} \succeq \left(\tilde{F}_{p-1}\tilde{F}_{p-1}^T - \frac{\Lambda \mathbf{1}_N \mathbf{1}_N^T \Lambda}{\|\mu\|_2^2} \right)^{\circ r} \quad (36)$$

D.2.7 • STEP 7: APPLICATION OF GERSHGORIN CIRCLE THEOREM

To bound the minimum eigenvalue of $(\tilde{F}_{p-1}\tilde{F}_{p-1}^T)^{\circ r}$, we apply the Gershgorin circle theorem:

$$(\tilde{F}_{p-1}^{*r})(\tilde{F}_{p-1}^{*r})^T \succeq \min_{i \in [N]} \|\tilde{F}_{p-1}^{(i)}\|_2^{2r} - N \max_{i \neq j} |\langle \tilde{F}_{p-1}^{(i)}, \tilde{F}_{p-1}^{(j)} \rangle|^r \quad (37)$$

Using concentration properties of centered features, we show:

$$\|\tilde{F}_{p-1}^{(i)}\|_2^{2r} = \Theta \left(\left(d \prod_{l=1}^{p-1} n_l \beta_l^2 \right)^r \right) \quad (38)$$

$$N \max_{i \neq j} |\langle \tilde{F}_{p-1}^{(i)}, \tilde{F}_{p-1}^{(j)} \rangle|^r = o \left(\left(d \prod_{l=1}^{p-1} n_l \beta_l^2 \right)^r \right) \quad (39)$$

with exponentially high probability.

D.2.8 • STEP 8: BOUNDING DERIVATIVE TERM NORMS, VERY TECHNICAL

To bound $\min_{i \in [N]} \|(B_{p+1})_{i:}\|_2^2$, we analyze each case:

For $p \in [L-2]$:

$$\|(B_{p+1})_{i:}\|_2^2 = \|\Sigma_{p+1}(x_i) \left(\prod_{l=p+2}^{L-1} W_l \Sigma_l(x_i) \right) W_L\|_2^2 \quad (40)$$

$$= \Theta \left(\beta_L^2 n_{p+1} \prod_{l=p+2}^{L-1} n_l \beta_l^2 \right) \quad (41)$$

This bound uses some very technical and important tools:

- **Operator norm for concentration of Gaussian matrices**
- **Inductive analysis of random matrix products for every layer**
- **Properties of derivative matrices $\Sigma_l(x)$ for every layer**

D.2.9 • STEP 9: FINAL ASSEMBLY AND WIDTH OPTIMIZATION

Combining all bounds via union bound yields:

$$\bar{K}^{(L)} \geq \sum_{p=0}^{L-1} \mu_r(\sigma)^2 \Theta \left(d \prod_{l=1}^p n_l \beta_l^2 \right) \Theta \left(\beta_L^2 \prod_{l=p+1}^{L-1} n_l \beta_l^2 \right) \quad (42)$$

$$= \mu_r(\sigma)^2 \Theta \left(d \beta_L^2 \sum_{p=0}^{L-1} \prod_{l=1}^{L-1} n_l \beta_l^2 \right) \quad (43)$$

The width condition requires at least one layer to satisfy:

$$n_p = \tilde{\Omega}(N \log N), \quad \text{with } \xi_p = 1 \quad (44)$$

where ξ_p indicates layer p is "wide".

D.2.10 • UPPER BOUND VIA DIAGONAL ANALYSIS

For the upper bound, we use:

$$JJ^T \leq (JJ^T)_{11} = \sum_{p=0}^{L-1} \|(F_p)_{1:}\|_2^2 \|(B_{p+1})_{1:}\|_2^2 \quad (45)$$

$$= \sum_{p=0}^{L-1} \|f_p(x_1)\|_2^2 \|(B_{p+1})_{1:}\|_2^2 \quad (46)$$

$$= \mathcal{O} \left(d \beta_L^2 \sum_{p=0}^{L-1} \prod_{l=1}^{L-1} n_l \beta_l^2 \right) \quad (47)$$

D.2.11 • FINAL RESULT

Assuming $\beta_l = 1$ for simplicity and that some layer p^* satisfies $n_{p^*} = \tilde{\Omega}(N)$, the main theorem gives:

$$\mu_r(\sigma)^2 \Omega(d) \leq \bar{K}^{(L)} \leq \mathcal{O}(dL) \quad (48)$$

with probability at least $1 - \text{poly}(N) \exp(-\Omega(\min_l n_l))$.

This approach significantly reduces width requirements compared to classical methods, requiring only one wide layer rather than all layers.

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